

Combinatorics PhD Exam, May 2011

1. Let $f(n)$ be the number of permutations of length n that have no cycle that is shorter than three.
 - (a) Find the exponential generating function of the sequence $f(n)$. You can assume that $f(0) = 1$.
 - (b) Find $\lim_{n \rightarrow \infty} f(n)/n!$.
2. We divide a group of n people into subgroups A , B , and C , and ask each subgroup to form a line. We also require that A have an odd number of people, and that B have an even number of people. How many ways are there to do this?
3. Let A be the graph obtained from K_n by deleting an edge. Find a formula for the number of spanning trees of A .
4. Let $X_n(p)$ be the number of cycles of the n -permutation p . Compute $\text{VAR}(X_n)$.
5. Show that a planar graph for which every face has an even number of edges must be bipartite.
6. Recall that a tournament is a complete directed graph. Prove that there exists a tournament on eight vertices that contains at least 316 Hamiltonian paths.
7. For which values of m and n is the complete bipartite graph $K_{m,n}$ planar, for which values of m and n is it Hamiltonian, and for which values of m and n is it Eulerian?
8. Prove in a finite poset P , the number of elements in the longest chain is equal to the smallest number k so that P can be decomposed into the union of k antichains.